

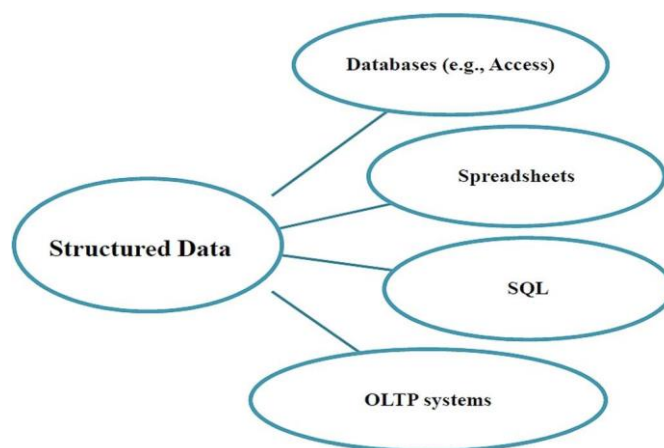
Big Data (Unit 1)

❖ Digital Data

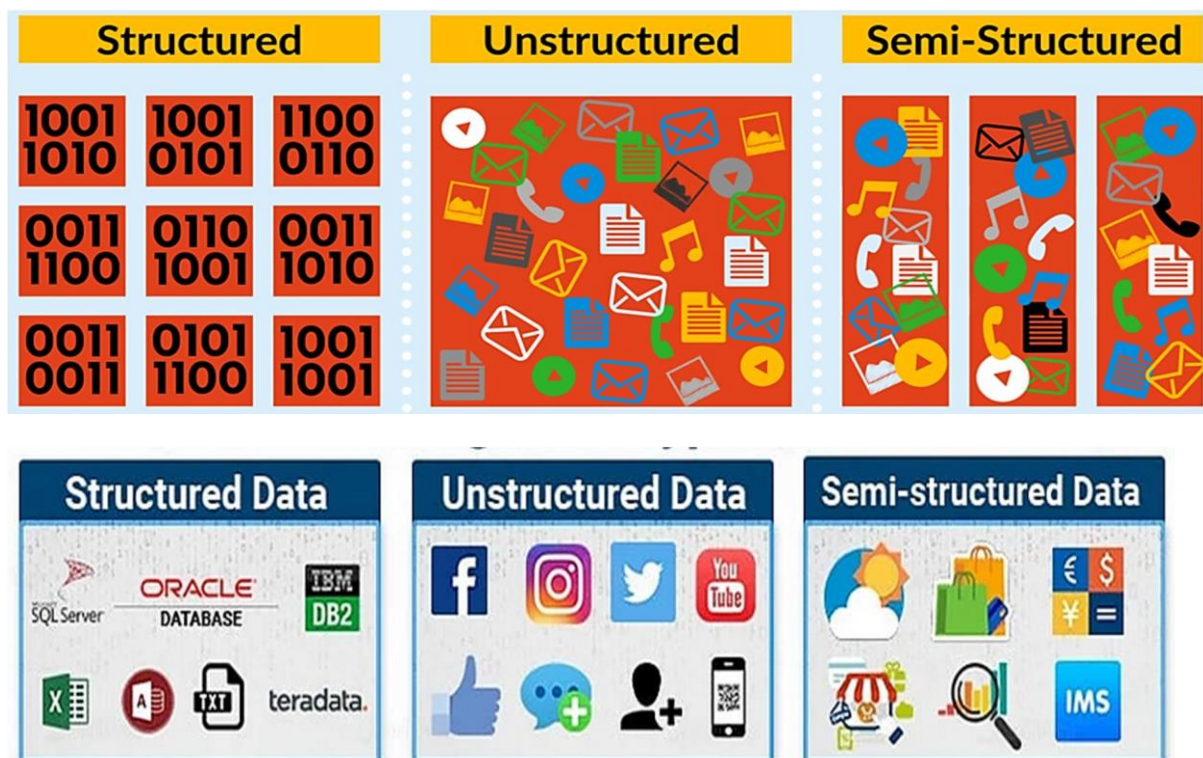
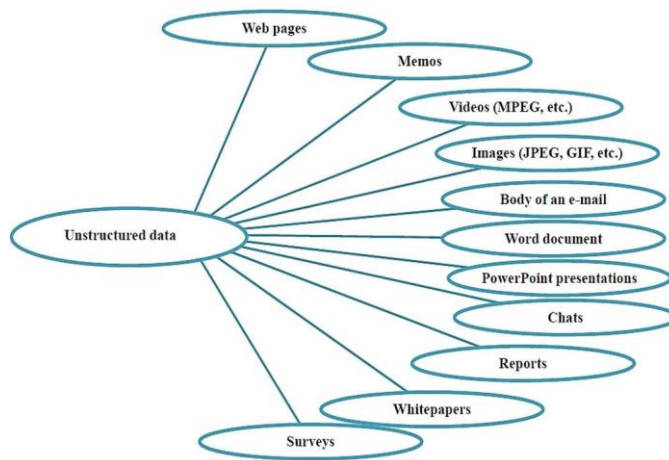
- In computing world, digital data is considered as a collection of facts that's transmitted and saved in a electronic format, and processed through software system.
- Digital data is data that represents other forms of data using specific machine language systems that can be interpreted by various technologies.
- Digital Data is generated by various devices, like desktops, laptops, tablets, mobile phones, and electronic sensors.
- Digital data is stored as strings of binary values (0s and 1s) on a storage medium that's either internal or external to the devices generating or accessing the information.
- For Example - Whenever you send an email, read a social media post, or take pictures with your digital camera, you are working with digital data.

❖ Types of Digital Data

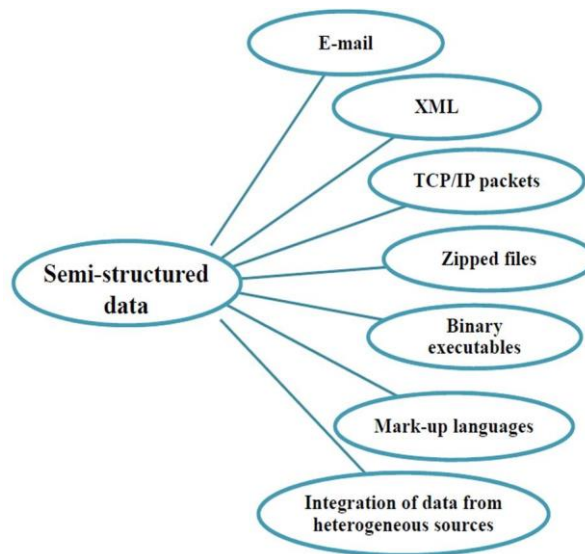
- **Structured Data :-** Structured data conforms to a data model or schema and is often stored in tabular form. It is used to capture relationships between different entities and is therefore most often stored in a relational database.
- Structured is one of the types of big data and By structured data, we mean data that can be processed, stored, and retrieved in a fixed format.
- This is the data which is in an organized form (e.g., in rows and columns) and can be easily used by a computer program.
- It refers to highly organized information that can be readily and seamlessly stored and accessed from a database by simple search engine algorithms.
- **For example** - the employee table in a company database will be structured as the employee details, their job positions, their salaries, etc., will be present in an organized manner.



- **Unstructured Data :-** Data that does not conform to a data model or data schema is known as unstructured data.
- Unstructured data refers to the data that lacks any specific form or structure whatsoever. This makes it very difficult and time-consuming to process and analyze unstructured data.
- About 80—90% data of an organization is in this format; for example, memos, chat rooms, PowerPoint presentations, images, videos, letters, researches, white papers, body of an email, etc.
- For example - To play a video file, it is essential that the correct codec (coder-decoder) is available. Unstructured data cannot be directly processed or queried using SQL. Email is an example of unstructured data.



- **Semi-Structured Data :-** Semi-structured data has a defined level of structure and consistency, but is not relational in nature. Instead, semi-structured data is hierarchical or graph-based. This kind of data is commonly stored in files that contain text.
- This is the data which does not conform to a data model but has some structure. Semi-structured data is information that does not reside in a relational database but that have some organizational properties that make it easier to analyze.
- It refers to highly organized information that can be readily and seamlessly stored and accessed from a database by simple search engine algorithms.
- **For example** - emails, XML, markup languages like HTML, etc.



❖ History of Big Data innovation

✓ Big Data phase 1.0

- Data analysis, data analytics and Big Data originate from the longstanding domain of database management. It relies heavily on the storage, extraction, and optimization techniques that are common in data that is stored in Relational Database Management Systems (RDBMS).
- Database management and data warehousing are considered the core components of Big Data Phase 1. It provides the foundation of modern data analysis as we know it today, using well-known techniques such as database queries, online analytical processing and standard reporting tools.

✓ Big Data phase 2.0

- Since the early 2000s, the Internet and the Web began to offer unique data collections and data analysis opportunities. With the expansion of web traffic and online stores, companies such as Yahoo, Amazon and eBay started to analyse customer behaviour by analysing click-rates, IP-specific location data and search logs. This opened a whole new world of possibilities.

- From a data analysis, data analytics, and Big Data point of view, HTTP-based web traffic introduced a massive increase in semi-structured and unstructured data. Besides the standard structured data types, organizations now needed to find new approaches and storage solutions to deal with these new data types in order to analyse them effectively. The arrival and growth of social media data greatly aggravated the need for tools, technologies and analytics techniques that were able to extract meaningful information out of this unstructured data.

✓ **Big Data phase 3.0**

- Although web-based unstructured content is still the main focus for many organizations in data analysis, data analytics, and big data, the current possibilities to retrieve valuable information are emerging out of mobile devices.
- Mobile devices not only give the possibility to analyze behavioral data (such as clicks and search queries), but also give the possibility to store and analyze location-based data (GPS-data). With the advancement of these mobile devices, it is possible to track movement, analyze physical behavior and even health-related data (number of steps you take per day). This data provides a whole new range of opportunities, from transportation, to city design and health care.
- Simultaneously, the rise of sensor-based internet-enabled devices is increasing the data generation like never before. Famously coined as the ‘Internet of Things’ (IoT), millions of TVs, thermostats, wearables and even refrigerators are now generating zettabytes of data every day. And the race to extract meaningful and valuable information out of these new data sources has only just begun.

BIG DATA PHASE 1	BIG DATA PHASE 2	BIG DATA PHASE 3
Period: 1970-2000	Period: 2000-2010	Period: 2010-present
DBMS-based, structured content: • RDBMS & data warehousing • Extract Transfer Load • Online Analytical Processing • Dashboards & scorecards • Data mining & statistical analysis	Web-based, unstructured content • Information retrieval and extraction • Opinion mining • Question answering • Web analytics and web intelligence • Social media analytics • Social network analysis • Spatial-temporal analysis	Mobile and sensor-based content • Location-aware analysis • Person-centered analysis • Context-relevant analysis • Mobile visualization • Human-Computer-Interaction

❖ **Introduction to Big Data Platform**

- Big data platform is a type of IT solution that combines the features and capabilities of several big data application and utilities within a single solution.
- It is an enterprise class IT platform that enables organization in developing, deploying, operating and managing a big data infrastructure /environment.

- Big data platform generally consists of big data storage, servers, database, big data management, business intelligence and other big data management utilities. It also supports custom development, querying and integration with other systems.
- The primary benefit behind a big data platform is to reduce the complexity of multiple vendors/ solutions into a one cohesive solution. Big data platform are also delivered through cloud where the provider provides an all inclusive big data solutions and services.

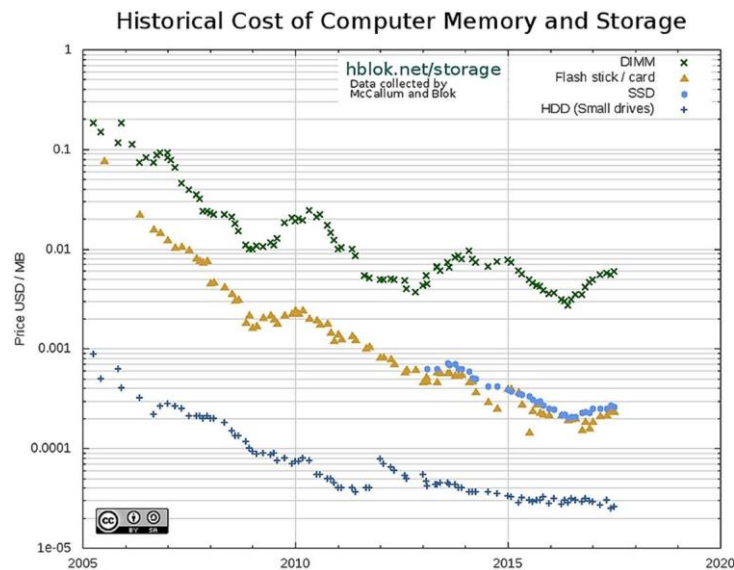
❖ **Characteristics of a Big Data Platform**

- Any good big data platform should have the following important features.
- ✓ Ability to accommodate new applications and tools depending on the evolving business needs.
- ✓ Support several data formats.
- ✓ Ability to accommodate large volumes of streaming or at-rest data.
- ✓ Have a wide variety of conversion tools to transform data to different preferred formats.
- ✓ Capacity to accommodate data at any speed.
- ✓ Provide the tools for scouring the data through massive data sets.
- ✓ Support linear scaling.
- ✓ The ability for quick deployment.
- ✓ Have the tools for data analysis and reporting requirements.

❖ **Drivers for Big Data**

- A number of business drivers are at the core of this success and six main business drivers can be identified:
- ✓ **1. The digitization of society:** - Big Data is largely consumer driven and consumer oriented. Most of the data in the world is generated by consumers, who are nowadays 'always-on'. Most people now spend 4-6 hours per day consuming and generating data through a variety of devices and (social) applications. With every click, swipe or message, new data is created in a database somewhere around the world. Because everyone now has a smartphone in their pocket, the data creation sums to incomprehensible amounts. Some studies estimate that 60% of data was generated within the last two years, which is a good indication of the rate with which society has digitized.
- ✓ **2. The plummeting of technology costs :-** Technology related to collecting and processing massive quantities of diverse (high variety) data has become increasingly more affordable. The costs of data storage and processors keep declining, making it possible for small businesses and individuals to become involved with Big Data. For storage capacity, the often-cited Moore's Law still holds that the storage density (and

therefore capacity) still doubles every two years. The plummeting of technology costs has been depicted in the figure below.



Besides the plummeting of the storage costs, a second key contributing factor to the affordability of Big Data has been the development of open source Big Data software frameworks. The most popular software framework (nowadays considered the standard for Big Data) is Apache Hadoop for distributed storage and processing. Due to the high availability of these software frameworks in open sources, it has become increasingly inexpensive to start Big Data projects in organizations.

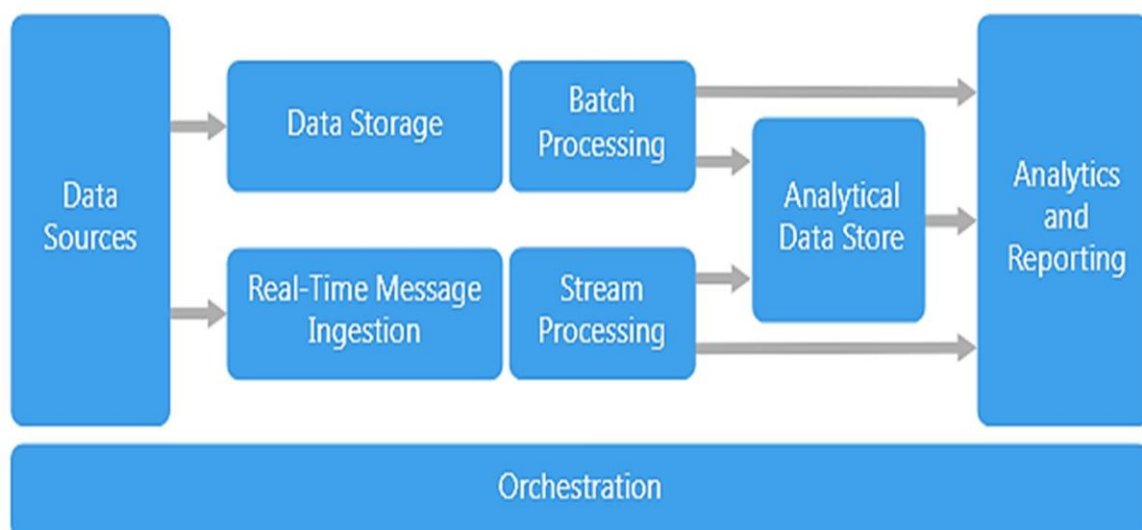
- ✓ **3. Connectivity through cloud computing :-** Cloud computing environments (where data is remotely stored in distributed storage systems) have made it possible to quickly scale up or scale down IT infrastructure and facilitate a pay-as-you-go model. This means that organizations that want to process massive quantities of data (and thus have large storage and processing requirements) do not have to invest in large quantities of IT infrastructure. Instead, they can license the storage and processing capacity they need and only pay for the amounts they actually used. As a result, most of Big Data solutions leverage the possibilities of cloud computing to deliver their solutions to enterprises.
- ✓ **4. Increased knowledge about data science :-** In the last decade, the term data science and data scientist have become tremendously popular. In October 2012, Harvard Business Review called the data scientist “sexiest job of the 21st century” and many other publications have featured this new job role in recent years. The demand for data scientist (and similar job titles) has increased tremendously and many people have actively become engaged in the domain of data science. As a result, the knowledge and education about data science has greatly professionalized and more information becomes available every day. While statistics and data analysis mostly remained an academic field previously, it is quickly becoming a popular subject among students and the working population.
- ✓ **5. Social media applications :-** Everyone understands the impact that social media has on daily life. However, in the study of Big Data, social media plays a role of paramount importance. Not only because of the sheer volume of data that is produced everyday through platforms such as Twitter, Facebook, LinkedIn and Instagram, but also because social media provides nearly real-time data about human behaviour.

Social media data provides insights into the behaviors, preferences and opinions of ‘the public’ on a scale that has never been known before. Due to this, it is immensely valuable to anyone who is able to derive meaning from these large quantities of data. Social media data can be used to identify customer preferences for product development, target new customers for future purchases, or even target potential voters in elections. Social media data might even be considered one of the most important business drivers of Big Data.

- ✓ **6. The upcoming internet of things (IoT) :-** The Internet of things (IoT) is the network of physical devices, vehicles, home appliances and other items embedded with electronics, software, sensors, actuators, and network connectivity which enables these objects to connect and exchange data. It is increasingly gaining popularity as consumer goods providers start including ‘smart’ sensors in household appliances. Whereas the average household in 2010 had around 10 devices that connected to the internet, this number is expected to rise to 50 per household by 2020. Examples of these devices include thermostats, smoke detectors, televisions, audio systems and even smart refrigerators.

❖ **Big Data Architecture**

A big data architecture is designed to handle the ingestion, processing, and analysis of data that is too large or complex for traditional database systems.



- ❖ **Data Sources :-** Data is sourced from multiple inputs in a variety of formats, including both structured and unstructured. Data sources, open and third-party, play a significant role in architecture. Relational databases, data warehouses, cloud-based data warehouses, SaaS applications, real-time data from company servers and sensors such as IoT devices, third-party data providers, and also static files such as Windows logs, comprise several data sources. Both batch processing and real-time processing are possible. The data managed can be both batch processing and real-time processing.
- ❖ **Data Storage:-** There is data stored in file stores that are distributed in nature and that can hold a variety of format-based big files. It is also possible to store large numbers of different format-based big files in the data lake. This consists of the data that is managed for batch built operations and is saved in the file stores. We provide HDFS, Microsoft Azure, AWS, and GCP storage, among other blob containers.
- ❖ **Batch Processing:-** Each chunk of data is split into different categories using long-running jobs, which filter and aggregate and also prepare data for analysis. These jobs typically require sources, process them, and deliver the processed files to new files.

Multiple approaches to batch processing are employed, including Hive jobs, U-SQL jobs, Sqoop or Pig and custom map reducer jobs written in any one of the Java or Scala or other languages such as Python.

- ❖ **Real-time message ingestion:-** If the solution includes real-time sources, the architecture must include a way to capture and store real-time messages for stream processing. This might be a simple data store, where incoming messages are dropped into a folder for processing. However, many solutions need a message ingestion store to act as a buffer for messages, and to support scale-out processing, reliable delivery, and other message queuing semantics. Options include Azure Event Hubs, Azure IoT Hubs, and Kafka.
- ❖ **Stream processing:-** After capturing real-time messages, the solution must process them by filtering, aggregating, and otherwise preparing the data for analysis. The processed stream data is then written to an output sink. Azure Stream Analytics provides a managed stream processing service based on perpetually running SQL queries that operate on unbounded streams. You can also use open source Apache streaming technologies like Storm and Spark Streaming in an HDInsight cluster.
- ❖ **Analytics-Based Data-store:-** In order to analyze and process already processed data, analytical tools use the data store that is based on HBase or any other NoSQL data warehouse technology. The data can be presented with the help of a hive database, which can provide metadata abstraction, or interactive use of a hive database, which can provide metadata abstraction in the data store. NoSQL databases like HBase or Spark SQL are also available.
- ❖ **Analysis and reporting:-** Most Big Data platforms are geared to extracting business insights from the stored data via analysis and reporting. This requires multiple tools. Structured data is relatively easy to handle, while more advanced and specialized techniques are required for unstructured data. Data scientists may undertake interactive data exploration using various notebooks and tool-sets. A data modeling layer might also be included in the architecture, which may also enable self-service BI using popular visualization and modeling techniques. Analytics results are sent to the reporting component, which replicates them to various output systems for human viewers, business processes, and applications. After visualization into reports or dashboards, the analytic results are used for data-driven business decision making.
- ❖ **Orchestration:-** Most big data solutions consist of repeated data processing operations, encapsulated in workflows, that transform source data, move data between multiple sources and sinks, load the processed data into an analytical data store, or push the results straight to a report or dashboard. To automate these workflows, you can use an orchestration technology such as Azure Data Factory or Apache Oozie and Sqoop.

❖ Big Data Characteristics & 5 Vs

Big Data contains a large amount of data that is not being processed by traditional data storage or the processing unit. It is used by many multinational companies to process the data and business of many organizations. The data flow would exceed 150 Exabyte's per day before replication. There are five V's of Big Data that explains the characteristics.

❖ 1. Volume:

- ✓ The name 'Big Data' itself is related to a size which is enormous.
- ✓ Volume is a huge amount of data.
- ✓ To determine the value of data, size of data plays a very crucial role. If the volume of data is very large then it is actually considered as a 'Big Data'. This means whether a particular data can actually be considered as a Big Data or not, is dependent upon the volume of data.
- ✓ Hence while dealing with Big Data it is necessary to consider a characteristic 'Volume'.

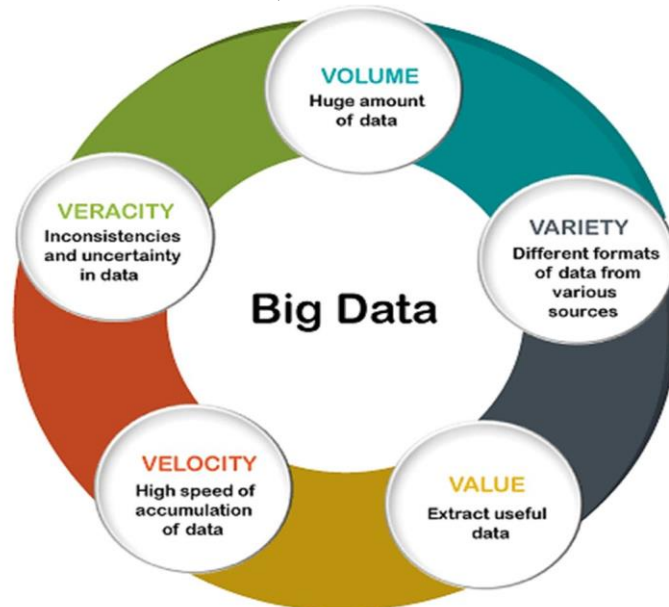
- ✓ **Example:** In the year 2016, the estimated global mobile traffic was 6.2 Exabytes(6.2 billion GB) per month. Also, by the year 2020 we will have almost 40000 ExaBytes of data.

❖ 2. Velocity:

- ✓ Velocity refers to the high speed of accumulation of data.
- ✓ In Big Data velocity data flows in from sources like machines, networks, social media, mobile phones etc.
- ✓ There is a massive and continuous flow of data. This determines the potential of data that how fast the data is generated and processed to meet the demands.
- ✓ Sampling data can help in dealing with the issue like 'velocity'.
- ✓ **Example:** There are more than 3.5 billion searches per day are made on Google. Also, FaceBook users are increasing by 22%(Approx.) year by year.

❖ 3. Variety:

- ✓ It refers to nature of data that is structured, semi-structured and unstructured data.
- ✓ It also refers to heterogeneous sources.
- ✓ Variety is basically the arrival of data from new sources that are both inside and outside of an enterprise. It can be structured, semi-structured and unstructured.



❖ 4. Veracity:

- ✓ It refers to inconsistencies and uncertainty in data, that is data which is available can sometimes get messy and quality and accuracy are difficult to control.
- ✓ Big Data is also variable because of the multitude of data dimensions resulting from multiple disparate data types and sources.
- ✓ Example: Data in bulk could create confusion whereas less amount of data could convey half or Incomplete Information.

❖ 5. Value:

- ✓ After having the 4 V's into account there comes one more V which stands for Value!. The bulk of Data having no Value is of no good to the company, unless you turn it into something useful.
 - Data in itself is of no use or importance but it needs to be converted into something valuable to extract Information.

❖ Big Data Technology Components

Given below are the main technology components of big data.

- **1. Machine Learning :-** It is the science of making computers learn stuff by themselves. In machine learning, a computer is expected to use algorithms and statistical models to perform specific tasks without any explicit instructions. Machine learning applications provide results based on past experience. For example, these days, there are some mobile applications that will give you a summary of your finances, bills, will remind you of your bill payments, and also may give you suggestions to go for some saving plans. These functions are done by reading your emails and text messages.
- **2. Natural Language Processing (NLP) :-** It is the ability of a computer to understand human language as spoken. The most obvious examples that people can relate to these days are google home and Amazon Alexa. Both use NLP and other technologies to give us a virtual assistant experience. NLP is all around us without us even realizing it. When writing a mail, while making any mistakes, it automatically corrects itself, and these days it gives auto-suggests for completing the mails and automatically intimidates us when we try to send an email without the attachment that we referenced in the text of the email, this is part of Natural Language Processing Applications which are running at the backend.
- **3. Business Intelligence :-** Business Intelligence (BI) is a method or process that is technology-driven to gain insights by analysing data and presenting it in a way that the end-users (usually high-level executives) like managers and corporate leaders can gain some actionable insights from it and make informed business decisions on it.
- **4. Cloud Computing :-** If we go by the name, it should be computing done on clouds; well, it is true, just here we are not talking about real clouds, cloud here is a reference for the Internet. So we can define cloud computing as the delivery of computing services servers, storage, databases, networking, software, analytics, intelligence, and more over the Internet (“the cloud”) to offer faster innovation, flexible resources, and economies of scale.

❖ Big Data Importance

Big Data importance doesn't revolve around the amount of data a company has. Its importance lies in the fact that how the company utilizes the gathered data. Every company uses its collected data in its own way. More effectively the company uses its data, more rapidly it grows. The companies in the present market need to collect it and analyze it because:

- ✓ **1. Cost Savings :-** Big Data tools like Apache Hadoop, Spark, etc. bring cost-saving benefits to businesses when they have to store large amounts of data. These tools help organizations in identifying more effective ways of doing business.
- ✓ **2. Time-Saving :-** Real-time in-memory analytics helps companies to collect data from various sources. Tools like Hadoop help them to analyse data immediately thus helping in making quick decisions based on the learning.
- ✓ **3. Understand the market conditions :-** Big Data analysis helps businesses to get a better understanding of market situations. For example, analysis of customer purchasing behavior helps companies to identify the products sold most and thus produces those products accordingly. This helps companies to get ahead of their competitors.
- ✓ **4. Social Media Listening :-** Companies can perform sentiment analysis using Big Data tools. These enable them to get feedback about their company, that is, who is saying what about the company.

- ✓ **5. Boost Customer Acquisition and Retention :-** Customers are a vital asset on which any business depends on. No single business can achieve its success without building a robust customer base. But even with a solid customer base, the companies can't ignore the competition in the market. If we don't know what our customers want then it will degrade companies' success. It will result in the loss of clientele which creates an adverse effect on business growth. Big data analytics helps businesses to identify customer related trends and patterns. Customer behavior analysis leads to a profitable business.
- ✓ **6. Solve Advertisers Problem and Offer Marketing Insights :-** Big data analytics shapes all business operations. It enables companies to fulfil customer expectations. Big data analytics helps in changing the company's product line. It ensures powerful marketing campaigns.
- ✓ **7. The driver of Innovations and Product Development :-** Big data makes companies capable to innovate and redevelop their products.

❖ Big Data Applications

Big data has found many applications in various fields today. The major fields where big data is being used are as follows.

- ✓ **Government :-** Big data analytics has proven to be very useful in the government sector. Big data analysis played a large role in Barack Obama's successful 2012 re-election campaign. Also most recently, Big data analysis was majorly responsible for the BJP and its allies to win a highly successful Indian General Election 2014. The Indian Government utilizes numerous techniques to ascertain how the Indian electorate is responding to government action, as well as ideas for policy augmentation.
- ✓ **Social Media Analytics :-** The advent of social media has led to an outburst of big data. Various solutions have been built in order to analyse social media activity like IBM's Cognos Consumer Insights, a point solution running on IBM's BigInsights Big Data platform, can make sense of the chatter. Social media can provide valuable real-time insights into how the market is responding to products and campaigns. With the help of these insights, the companies can adjust their pricing, promotion, and campaign placements accordingly. Before utilizing the big data there needs to be some pre-processing to be done on the big data in order to derive some intelligent and valuable results. Thus to know the consumer mind-set the application of intelligent decisions derived from big data is necessary.
- ✓ **Technology :-** The technological applications of big data comprise of the following companies which deal with huge amounts of data every day and put them to use for business decisions as well. For example, eBay.com uses two data warehouses at 7.5 petabytes and 40PB as well as a 40PB Hadoop cluster for search, consumer recommendations, and merchandising. Inside eBay's 90PB data warehouse. Amazon.com handles millions of back-end operations every day, as well as queries from more than half a million third-party sellers. The core technology that keeps Amazon running is Linux-based and as of 2005, they had the world's three largest Linux databases, with capacities of 7.8 TB, 18.5 TB, and 24.7 TB. Facebook handles 50 billion photos from its user base. Windermere Real Estate uses anonymous GPS signals from nearly 100 million drivers to help new home buyers determine their typical drive times to and from work throughout various times of the day.
- ✓ **Fraud Detection :-** For businesses whose operations involve any type of claims or transaction processing, fraud detection is one of the most compelling Big Data application examples. Historically, fraud detection on the fly has proven an elusive goal. In most cases, fraud is discovered long after the fact, at which point the damage has been

done and all that's left is to minimize the harm and adjust policies to prevent it from happening again. Big Data platforms that can analyze claims and transactions in real time, identifying large-scale patterns across many transactions or detecting anomalous behavior from an individual user, can change the fraud detection game.

- ✓ **Call Center Analytics :-** Now we turn to the customer-facing Big Data application examples, of which call center analytics are particularly powerful. What's going on in a customer's call center is often a great barometer and influencer of market sentiment, but without a Big Data solution, much of the insight that a call center can provide will be overlooked or discovered too late. Big Data solutions can help identify recurring problems or customer and staff behavior patterns on the fly not only by making sense of time/quality resolution metrics but also by capturing and processing call content itself.
- ✓ **Banking :-** The use of customer data invariably raises privacy issues. By uncovering hidden connections between seemingly unrelated pieces of data, big data analytics could potentially reveal sensitive personal information. Research indicates that 62% of bankers are cautious in their use of big data due to privacy issues. Further, outsourcing of data analysis activities or distribution of customer data across departments for the generation of richer insights also amplifies security risks. Such as customers' earnings, savings, mortgages, and insurance policies ended up in the wrong hands. Such incidents reinforce concerns about data privacy and discourage customers from sharing personal information in exchange for customized offers.
- ✓ **Agriculture :-** A biotechnology firm uses sensor data to optimize crop efficiency. It plants test crops and runs simulations to measure how plants react to various changes in condition. Its data environment constantly adjusts to changes in the attributes of various data it collects, including temperature, water levels, soil composition, growth, output, and gene sequencing of each plant in the test bed. These simulations allow it to discover the optimal environmental conditions for specific gene types.
- ✓ **Marketing :-** Marketers have begun to use facial recognition software to learn how well their advertising succeeds or fails at stimulating interest in their products. A recent study published in the Harvard Business Review looked at what kinds of advertisements compelled viewers to continue watching and what turned viewers off. Among their tools was "a system that analyses facial expressions to reveal what viewers are feeling." The research was designed to discover what kinds of promotions induced watchers to share the ads with their social network, helping marketers create ads most likely to "go viral" and improve sales.
- ✓ **Smart Phones :-** Perhaps more impressive, people now carry facial recognition technology in their pockets. Users of I Phone and Android smartphones have applications at their fingertips that use facial recognition technology for various tasks. For example, Android users with the remember app can snap a photo of someone, then bring up stored information about that person based on their image when their own memory lets them down a potential boon for salespeople.
- ✓ **Telecom :-** Now a day's big data is used in different fields. In telecom also it plays a very good role. Operators face an uphill challenge when they need to deliver new, compelling, revenue-generating services without overloading their networks and keeping their running costs under control. The market demands new set of data management and analysis capabilities that can help service providers make accurate decisions by taking into account customer, network context and other critical aspects of their businesses. Most of these decisions must be made in real time, placing additional pressure on the operators. Real-time predictive analytics can help leverage the data that resides in their multitude systems, make it immediately accessible and help correlate that data to generate insight that can help them drive their business forward.
- ✓ **Healthcare :-** Traditionally, the healthcare industry has lagged behind other industries in the use of big data, part of the problem stems from resistance to change providers are accustomed to making treatment decisions independently, using their own clinical

judgment, rather than relying on protocols based on big data. Other obstacles are more structural in nature. This is one of the best place to set an example for Big Data Application. Even within a single hospital, payor, or pharmaceutical company, important information often remains siloed within one group or department because organizations lack procedures for integrating data and communicating findings. Health care stakeholders now have access to promising new threads of knowledge. This information is a form of “big data,” so called not only for its sheer volume but for its complexity, diversity, and timelines. Pharmaceutical industry experts, payers, and providers are now beginning to analyse big data to obtain insights. Recent technologic advances in the industry have improved their ability to work with such data, even though the files are enormous and often have different database structures and technical characteristics.

❖ **Big Data Features**

- **1). Easy Result Formats :-** Results are imperative parts of big data analytics model as they support in the decision-making process, that are made to decide future strategy and goals. Scientists prefer the results to get the result in the real-time so that they can take better and appropriate decisions, based on the analysis result. The tools must be able to produce a result in such a way that it can provide insights into data analysis and decision-making platform. The platform should be able to provide the real-time streams that can help in making instant and quick decisions.
- **2). Raw data Processing :-** Here, the data processing means collecting and organizing data in a meaningful manner. Data modeling takes complex data sets and displays them in the visual form or diagram or chart. Here, data should be interpretable and digestible so that it can be used in making decisions. Tools of big data analytics must be able to import data from various data sources such as Microsoft Access, text files, Microsoft Excel and other files. Tools must be able to collect data from multiple data sources and in multiple formats. In this way need for data conversion will be reduced and overall process speed will be improved. Even the export quality and capability to visualize data sets and handling various formats like PDFs, Excel, or Word files can be used directly to collect and transfer the data. Below-listed features are essential for the data processing tools:
 - ✓ Data Mining
 - ✓ Data Modeling
 - ✓ File Exporting
 - ✓ Data File Sources
- **3). Prediction apps or Identity Management :-** Identity management is also a required and essential feature for any data analytics tool. The tool should be able to access any system and all related information that may be related to the computer hardware, software or any other individual computer. Here, the identity management system is also related to managing all issues related to the identity, data protection, and access so that it can support system, network passwords, and protocols. Here, it should be clear that whether a user can access the system or not and to which level the system access permission is granted? Identity management applications and system ensure that only authenticated users can access the system information and the tool or system must be able to organize a security plan and include fraud analytics and real-time security.
- **4). Reporting Feature :-** Businesses remain on top with the help of reporting features. Even time-to-time data should be fetched and represented in a well-organized manner. These way decision-makers can take timely decisions and handle the critical situations as well, especially in a society that is moving rapidly. Data tools use dashboards to present KPIs and metrics. The reports must be customizable and target data set oriented. The expected capabilities of reporting tools are Real-time reporting, dashboard management, and location-based insights.

- **5). Security Features :-** For any successful business, it is essential to save their data. The tools that are used for big data analytics should offer safety and security to the data. For this there should be SSO feature that is known as a single sign-on feature with the help of that there is no need for the user to sign-in multiple times during the same session, even with the help of single or same login user can log in multiple times and monitor user activities and accounts. Moreover, data encryption is also an imperative feature that should be provided by Big Data analytics tools. It means to change the form of data or to make it unreadable from a readable form by using several algorithms and codes. Sometimes automatic encryption is also offered by web browsers. Comprehensive encryption capabilities are also offered by data analytics tools. For this single sign-on and data encryption are two of the most used and popular features.
- **6). Fraud management :-** A variety of fraud detection functionalities remain involved in the fraud analytics. Mainly when it comes to the fraud detection activities then it involves various fraud analytics. Due to these activities, businesses mainly focus on the way with which they will deal with the fraud rather than preventing any fraud. Fraud detection can be performed by data analytics tools. The tools should be able to perform repeated tests on the data at any time just to ensure that there will be no amiss. In this way, threats can be identified quickly and efficiently. With effective fraud analytics and identity management capabilities.
- **7). Technologies Support :-** Your data analytics tool must support the latest tools and technologies, especially those that are important for your organization. Here, one most important one is the A/B testing that is also known as the bucket or split testing, in this testing two webpage versions are compared to determine the performance of a better page. Here both the versions are compared on the basis in which user interacts with the webpage and then the best one is considered. Moreover, as far as technical support is concerned then your tool must be able to integrate with Hadoop, that is a set of open-source programs that can work as the backbone of data-analytics activities. Hadoop mainly involves the following four modules with which integration is expected:
 - ✓ **MapReduce:** It can read data from a file system that can be interpreted in the visualized manner.
 - ✓ **Hadoop Common:** For this, Java tool collection may be required to read data stored in the user's file system.
 - ✓ **YARN:** It is responsible to manage system resources so that data can be stored and analysis can be performed
 - ✓ **Distributed File System:** It allows data to be stored in an easy format. If the results of tools will be integrated with these Hadoop modules then the user can easily send the results to the user system. In this way flexibility, interoperability and both way communication can be ensured between organizations.
- **8). Version Control :-** Most of the data analytics tools are involved in adjusting data analytics model parameters. But it may cause problems when pushed into production. Version control feature of big analytics tools will surely improve the capabilities to track changes and it is able to release previous versions too whenever needed.
- **9). Scalability :-** Data will not the same all the times but it will grow as your organization is growing. With big data tools, this is always easy to scale-up as soon as new data is collected for the company and it can be analyzed well as expected. Also, the meaningful insights driven from data is pushed or integrated into the previous data successfully.
- **10). Quick Integrations :-** With integration capabilities, this is always easy to share data results with developers and data scientists. Big data tools always support the quick integration with cloud apps, data warehouses, other databases etc.

❖ Big Data Security & steps of securing Big Data

Big data security is the collective term for all the measures and tools used to guard both the data and analytics processes from attacks, theft, or other malicious activities that could harm or negatively affect them. Big Data can come in a mix of structured formats (organized into rows and columns containing numbers, dates, etc) or unstructured (social media data, PDF files, emails, images, etc).

- **1. Cloud Security Monitoring :-** Cloud computing generally offers more efficient communication and increased profitability for all businesses. This communication needs to be secure. Big data security offers cloud application monitoring. This provides host sensitive data and also monitors cloud-hosted infrastructure. Solutions also offer support across several relevant cloud platforms.
- **2. Network Traffic Analysis :-** Traffic continually moves in and out of your network. Due to the high volume of data over the network, it is difficult to maintain transactional visibility over the network traffic. Security analytics allow your enterprise to watch over this network traffic. It is used to establish baselines and detect anomalies. This also helps in cloud security monitoring. It is used to analyze traffic in and out of cloud infrastructure. It also illuminates dark spaces that are hidden in infrastructures and analyze encrypted sensitive data. Thus, ensuring the proper working of channels.
- **3. Insider Threat Detection :-** Insider threats are as much as a danger to your enterprise as external threats. An active malicious user can do as much damage as any malware attack. But it is only in some rare cases that an insider threat can destroy a network.
- **4. Threat Hunting :-** Generally, the IT Security team mostly engage in threat hunting. They search for potential indicators of dwelling threats and breaches that try to attack the IT infrastructure. Security analytics helps to automate this threat of hunting. It acts as an extra set of eyes for your threat hunting efforts. Threats hunting automation can help in detecting malware beaconing activity and thus alerts for its stoppage as soon as possible.
- **5. Incident Investigation :-** Generally, the sheer number of security alerts from SIEM solutions would overwhelm your IT security team. These continuous alerts can cause more fostering burnout and frustration. Thus to minimize this issue, security analytics automates the incident investigation by providing contextualization to alerts. Thus your team has more time to prioritize incidents and can deal with potential breach incidents first.
- **6. User Behaviour Analysis :-** Organization's users generally interact with your IT infrastructure all the time. Mainly it is the user's behavior that decides the success or failure of your cybersecurity. Therefore there is a need for tracking user's behavior. The security analytics monitor the unusual behavior of employees. Thus it helps to detect an insider threat or a malicious account. It can also detect suspicious patterns by correlating malicious activities. An example of one such renowned security analytics use case is UEBA. It helps to provide visibility into the IT environment. Thus compiling user activities from multiple datasets into complete profiles.
- **7. Data Exfiltration Detection :-** Data exfiltration is termed as any unauthorized movement of data moving in and out of any network. Unauthorized data movements can cause theft and leakage of data. Thus there is a need to protect data from such unauthorized access. Security analytics helps to detect the data exfiltration over a network. It is generally used to detect data leakage in encrypted communications.
- **8. Advertisement :-** With the help of security analytics, organizations can easily detect the insider threats. This is anticipated through behaviors such as abnormal login times, unusual email usage, and unauthorized database access requests. Sometimes it also looks for indicators that ask for visibility to third-party actors.

❖ Big data ethics

The field of big data ethics itself is defined as outlining, defending and recommending concepts of right and wrong practice when it comes to the use of data, with particular emphasis on personal data. Big data ethics aims to create an ethical and moral code of conduct for data use. There are five main areas of concern in big data ethics that outline the potential for immoral use of data:

1. Informed Consent

To consent means that you give uncoerced permission for something to happen to you. Informed consent is the most careful, respectful and ethical form of consent. It requires the data collector to make a significant effort to give participants a reasonable and accurate understanding of how their data will be used. In the past, informed consent for data collection was typically taken for participation in a single study. Big data makes this form of consent impossible as the entire purpose of big data studies, mining and analytics is to reveal patterns and trends between data points that were previously inconceivable. In this way, consent cannot possibly be 'informed' as neither the data collector nor the study participant can reasonably know or understand what will be garnered from the data or how it will be used. Revisions to the standard of informed consent have been introduced. The first is known as 'broad consent', which pre-authorises secondary uses of data. The second is 'tiered consent', which gives consent to specific secondary uses of data, for example, for cancer research but not for genomic research. Some argue that these newer forms of consent are a watering down of the concept and leave users open to unethical practices.

2. Privacy

The ethics of privacy involve many different concepts such as liberty, autonomy, security, and in a more modern sense, data protection and data exposure. You can understand the concept of big data privacy by breaking it down into three categories:

- ✓ **The condition of privacy**
- ✓ **The right to privacy**
- ✓ **The loss of privacy and invasion**

The scale and velocity of big data pose a serious concern as many traditional privacy processes cannot protect sensitive data, which has led to an exponential increase in cybercrime and data leaks.

One example of a significant data leak that caused a loss of privacy to over 200 million internet users happened in January 2021. A rising Chinese social media site called Sociallarks suffered a breach due to a series of data protection errors that included an unsecured ElasticSearch database. A hacker was able to access and scrape the database which stored:

- Names
- Phone numbers
- Email addresses
- Profile descriptions
- Follower and engagement data
- Locations
- LinkedIn profile links
- Connected social media account login names

A further concern is the growing analytical power of big data, i.e. how this can impact privacy when personal information from various digital platforms can be mined to create a full picture of a person without their explicit consent. For example, if someone applies for a job, information can be gained about them via their digital data footprint to identify political leanings, sexual orientation, social life, etc. All of this data could be used as a reason to reject an employment application even though the information was not offered up for judgement by the applicant.

3. Ownership

When we talk about ownership in big data terms, we steer away from the traditional or legal understanding of the word as the exclusive right to use, possess, and dispose of property. Rather, in this context, ownership refers to the redistribution of data, the modification of data, and the ability to benefit from data innovations.

In the past, legislators have ruled that as data is not property or a commodity, it, therefore, cannot be stolen - this belief offers little protection or compensation to internet users and consumers who provide valuable information to companies without personal benefit.

We can split ownership of data into two categories:

- The right to **control** data - edit, manage, share and delete data
- The right to **benefit** from data - profit from the use or sale of data

Contrary to common belief, those who generate data, for example, Facebook users, do not automatically own the data. Some even argue that the data we provide to use 'free' online platforms is in fact a payment for that platform. But big data is big money in today's world. Many internet users feel that the current balance is tilted against them when it comes to ownership of data and the transparency of companies who use and profit from the data we share.

In recent years, the idea of monetising personal data has gained traction; the ideology aims to give ownership of data back to the user and balance the market by allowing users to sell their data legally. This is a highly contentious field of legislation, and some argue that to designate data as a commodity is to lose our autonomy and freedoms.

4. Algorithm bias and objectivity

Algorithms are designed by humans, the data sets they study are selected and prepared by humans, and humans have bias. So far, there is significant evidence to suggest that human prejudices are infecting technology and algorithms, and negatively impacting the lives and freedoms of humans. Particularly those who exist within the minorities of our societies.

The so-called "coded bias" has been identified in such high-profile cases as MIT lab researcher [Joy Buolamwini's](#) discovery of racial skin-type bias from commercial artificial intelligence systems created by giant US companies. Buolamwini found that the software had been trained on datasets of 77% male pictures and more than 83% white-skinned pictures. These biased datasets created a situation wherein the program fails to recognise white male faces at an error rate of only 0.8%, whereas dark-skinned female faces are detected at an error rate of 20% in one case and 34% in the other two. These biases extend beyond racial and gendered lines and into the issues of criminal profiling, poverty and housing.

Algorithm biases have become such an ingrained part of everyday life that they have also been documented as impacting our personal psyches and thought processes. The phenomenon occurs when we perceive our reality to be a reflection of what we see online. However, what we view is often a tailored reality created by algorithms and personalised using our previous viewing habits. The algorithm shows us content that we are most likely to enjoy or agree with and discards the rest. When filter bubbles like this exist they create echo chambers and, in extreme cases, can lead to radicalisation, sectarianism and social isolation.

5. Big data divide

The big data divide seeks to define the current state of data access; the understanding and mining capabilities of big data is isolated within the hands of a few major corporations. These divides create ‘haves’ and ‘have nots’ in big data and exclude those who lack the necessary financial, educational and technological resources to access and analyse big datasets.

[Tim Berners-Lee](#) has argued that the big data divide separates individuals from data that could be highly valuable to their wellbeing. And despite the growing industry of applications that use data to enhance our lives in terms of health, finance, etc., there is currently no way for individuals to mine their own data or connect potential data silos missed by commercial software. Again, we face the ethical problem of who owns the data we generate; if our data is not ours to modify, analyse and benefit from on our own terms, then indeed we do not own it.

The data divide creates further problems when we consider algorithm biases that place individuals in categories based on a culmination of data that individuals themselves cannot access. For example, profiling software can mark a person as a high-risk potential for committing criminal activity, causing them to be legally stop-and-searched by authorities or even denied housing in certain areas. The big data divide means that the ‘data poor’ cannot understand the data or methods used to make these decisions about them and their lives.

❖ Big Data Compliance

Big data affects the compliance process directly because you will be expected to account for its flow inside your organization. The regulatory bodies are keen to examine every stage of data handling, including the collection, processing, and storage of data. The primary reason for the comprehensive evaluation is to make sure that the data is safe from cyberattacks. In order to get compliance status, you will build security measures to secure the data. During the analysis, you are expected to show how each of the techniques for risk mitigation works and their level of effectiveness. This thorough report on the data protection programs will make the organization’s certification easier.

Unlike organizations dealing with small data, using sophisticated analytics tools will be needed. You must also employ qualified professionals to analyze the data, identify security threats, and suggest strategies for mitigating them. During the enforcement process, managing big data would take more resources compared to handling small data. However, organizations will take advantage of managing big data, which helps to get direct predictive information about the probability of an attack. In auditing, the auditors are likely to adopt more rigorous steps when using this type of data than when using small data. As such, the use of big data analytics is one of the surest ways of building some of the organization’s most robust security systems. How Big Data is used in the

Process of Compliance. Big data assists the creation of a compressive risk assessment framework by:

- **Fraudulent Crime Prevention:** The use of big data strengthens the approach to predictive analysis, which is an effective way of detecting criminal activities such as money laundering. If a compliance officer uses big data for internal audits, cyber risks are discovered, and they intervene to avoid their occurrence. It speeds up the process of compliance and builds trust among your clients.
- **Managing Third Parties Threat:** If you are in the process of obtaining compliance certifications, you must maintain the risk associated with sharing the data with vendors appropriately. Big data analytics can help you manage vendor-related risks. This you will accomplish by carefully evaluating their ability to protect your data before sharing with them.
- **Helps in Customer Service:** You are required to prove that your customers are pleased with how you treat their data before you get any compliance certification. If you apply big data analytics, you will understand your customer's behavior, which will directly influence the decision-making process, thereby enabling the compliance process.

❖ Big Data Auditing

Many stakeholders don't view the auditing process as applicable during big data analytics. However, real-time data analysis can be applied in many different ways to benefit auditors. Auditors can use big data to expand the scope of their projects and draw comparisons over larger populations of data. Because big data involves the use of automation and artificial intelligence, data can be processed in larger volumes and higher velocity to uncover valuable insights for auditors. For example, previous cases of non-compliance, current policy changes, and fraud can be identified and used to guide the focus of both internal and external auditors. Big data is also helping financial auditors to streamline the reporting process and detect fraud. These professionals can identify business risks in time and conduct more relevant and accurate audits. Even before you can take advantage of analytics during auditing, you need adequate data management and aggregation processes in place. All data sets being analyzed for the purpose of auditing first need to be checked for accuracy, timeliness, and capacity. In this way, the downstream decisions that auditors make are based on reliable and high-quality information. This is particularly important when it comes to auditing because decisions regarding compliance, risk, and investment are typically made after audit reports are prepared. Big data also allows you to automate multiple portions of the auditing process. Human error is a common reason why businesses fall out of compliance or spend too much on audit-related requirements. By automating manual and repetitive tasks, auditors can set up various controls in advance and monitor how well a company is adhering to established guidelines.

❖ Big Data Protection

Data protection is a set of strategies and processes you can use to secure the privacy, availability, and integrity of your data. It is sometimes also called data security. A data protection strategy is vital for any organization that collects, handles, or stores sensitive data. A successful strategy can help prevent data loss, theft, or corruption and can help minimize damage caused in the event of a breach or disaster.

What Are Data Protection Principles?

Data protection principles help protect data and make it available under any circumstances. It covers operational data backup and business continuity/disaster recovery (BCDR) and involves implementing aspects of data management and data availability.

Here are key data management aspects relevant to data protection:

- **Data availability**—ensuring users can access and use the data required to perform business even when this data is lost or damaged.
- **Data lifecycle management**—involves automating the transmission of critical data to offline and online storage.
- **Information lifecycle management**—involves the valuation, cataloging, and protection of information assets from various sources, including facility outages and disruptions, application and user errors, machine failure, and malware and virus attacks.

What Is Data Privacy and Why Is It Important?

Data privacy is a guideline for how data should be collected or handled, based on its sensitivity and importance. Data privacy is typically applied to personal health information (PHI) and personally identifiable information (PII). This includes financial information, medical records, social security or ID numbers, names, birthdates, and contact information.

Data privacy concerns apply to all sensitive information that organizations handle, including that of customers, shareholders, and employees. Often, this information plays a vital role in business operations, development, and finances.

Data privacy helps ensure that sensitive data is only accessible to approved parties. It prevents criminals from being able to maliciously use data and helps ensure that organizations meet regulatory requirements.

❖ Big Data Ethics

Big data ethics also known as simply data ethics refers to systemizing, defending, and recommending concepts of right and wrong conduct in relation to data, in particular personal data. Since the dawn of the Internet the sheer quantity and quality of data has dramatically increased and is continuing to do so exponentially. Big data describes this large amount of data that is so voluminous and complex that traditional data processing application software is inadequate to deal with them. Recent innovations in medical research and healthcare, such as high-throughput genome sequencing, high-resolution imaging, electronic medical patient records and a plethora of internet-connected health devices have triggered a data deluge that will reach the exabyte range in the near future. Data ethics is of increasing relevance as the quantity of data increases because of the scale of the impact.

Big data ethics serves as a branch of ethics that evaluates data practices by collecting, generating, analyzing, and distributing data. As the world expands its digital footprint, data collected has the potential to impact people and thus society. With big data scandals left and right, users are voicing their concerns about their online privacy. Companies must adhere to a data ethics framework to maintain the trust of existing and future customers as well as business partners. A framework exists to provide transparency, so

the public knows what data you collect and why. People everywhere want to feel reassured that their data doesn't fall into the wrong hands. Much of this data consists of Personally identifiable information (PII):

- ✓ Full name
- ✓ Birthdate
- ✓ Street address
- ✓ Phone number
- ✓ Social security number
- ✓ Credit card information
- ✓ Bank account information
- ✓ Passport number

- **Why is Data ethics important?**

Once an organization fails to act ethically, it's no secret that it damages the company's brand and reputation. Similarly, after the many data scandals that occurred in the past couple of years, people lost trust in companies that manipulated customers'. However, these scandals don't just consist of data manipulation and sale. Housing data and keeping it safe from harm's way also is part of big data ethics. Some of the top data breaches ever to occur had lasting effects on brand trustworthiness. Therefore, adopting a concrete big data ethics framework is essential for the success of any large organization. Companies must act as information protectors as long as they choose to collect it.

- ✓ *Ethical Issues in the Big Data Industry*

The bigger the data central to the business, the higher its risk of violating customer privacy and individual rights. In 2022, the responsibility to actively manage data privacy and security falls on roles within the large organization.

- ✓ *Privacy*

When users submit their information, it's with the expectation that companies will keep it to themselves. Two common scenarios exist when this information is no longer private:

- A data breach
- A sale of information to a third party

- ✓ *Cybersecurity is a growing field.*

With its growth, users expect talented IT professionals to be able to protect their data. If a data breach occurs, the company fails to meet privacy expectations. Furthermore, in the 21st century, consumers expect large companies to have the means to protect data if they choose to collect it.

- ✓ *Lack of transparency*

Many users are unaware their information is being collected. In addition to being unaware, companies go to lengths to make it very inconspicuous that they do so. Websites add cookie opt-ins on pop-ups so that the user will accept to quickly see the page. After getting a user to submit information, some companies don't disclose how they use a person's data. Long lists of legal documentation are formatted in a way no

user expects to read through it. It's only after scandals or some type of media reporting do people discover the company's data collection method is unsatisfactory.

✓ *Lack of governance*

Before big data, the method to collect information was simple. People either gave you a physical copy of their information or they didn't. Companies stored the physical files under lock and key. Although someone could always potentially steal someone's identity, criminals had difficulty doing so to masses of people at one time. Now in a new age of information abundance, we have users unknowingly submitting heaps of information. The possibilities of using that information with AI and algorithms to someone's advantage are endless. In some countries, politicians are now creating laws to hinder certain actions. However, because big data collection is still relatively new, many laws don't exist.

✓ *Bias and discrimination*

Algorithms make assumptions about users. Depending on the assumption, they can begin to discriminate. For example, court systems started using algorithms to evaluate the criminal risk potential defendants and have used this data while sentencing. If the data encompasses a certain gender, nationality, or race then the results house bias against groupings outside of those specific groups.

✓ *Big Data Ethics Framework and Other Ways to Resolve Ethical Issues*

The government and large businesses now create a big data ethics framework to avoid ethical issues in the big data industry. While receiving initial consent, a company should develop competencies that voice how they use the data in an easily digestible manner. Most businesses have a set of mission statements and values. Now, many will also house a big data ethics framework as well.

❖ **Big Data privacy**

Big data privacy involves properly managing big data to minimize risk and protect sensitive data. Because big data comprises large and complex data sets, many traditional privacy processes cannot handle the scale and velocity required. To safeguard big data and ensure it can be used for analytics, you need to create a framework for privacy protection that can handle the volume, velocity, variety, and value of big data as it is moved between environments, processed, analyzed, and shared.

- **Big data includes big privacy concerns**

In an era of multi-cloud computing, data owners must keep up with both the pace of data growth and the proliferation of regulations that govern it especially regulations protecting the privacy of sensitive data and personally identifiable information (PII). With more data spread across more locations, the business risk of a privacy breach has never been higher, and with it, consequences ranging from high fines to loss of market share. Big data privacy is also a matter of customer trust. The more data you collect about users, the easier it gets to "connect the dots:" to understand their current behavior, draw inferences about their future behavior, and eventually develop deep and detailed profiles of their lives and preferences. The more data you collect, the more important it is to be transparent with your customers about what you're doing with their data, how you're storing it, and what steps you're taking

to comply with regulations that govern privacy and data protection. The volume and velocity of data from existing sources, such as legacy applications and e-commerce, is expanding fast. You also have new (and growing) varieties of data types and sources, such as social networks and IoT device streams. To keep pace, your big data privacy strategy needs to expand, too.

▪ **Predictions for big data privacy: What to expect**

- ✓ Prediction 1: Data privacy mandates will become more common.

As organizations store more types of sensitive data in larger amounts over longer periods of time, they will be under increasing pressure to be transparent about what data they collect, how they analyze and use it, and why they need to retain it. The European Union's General Data Privacy Regulation (GDPR) is a high-profile example. More government agencies and regulatory organizations are following suit. To respond to these growing demands, companies need reliable, scalable big data privacy tools that encourage and help people to access, review, correct, anonymize, and even purge some or all of their personal and sensitive information.

- ✓ Prediction 2: New big data analytic tools will enable organizations to perform deeper analysis of legacy data, discover uses for which the data wasn't originally intended, and combine it with new data sources. Big data analytics tools and solutions can now dig into data sources that were previously unavailable, and identify new relationships hidden in legacy data. That's a great advantage when it comes to getting a complete view of your enterprise data especially for customer 360 and analytics initiatives. But it also raises questions about the accuracy of aging data and the ability to track down entities for consent to use their information in new ways.

❖ **Big Data Analytics**

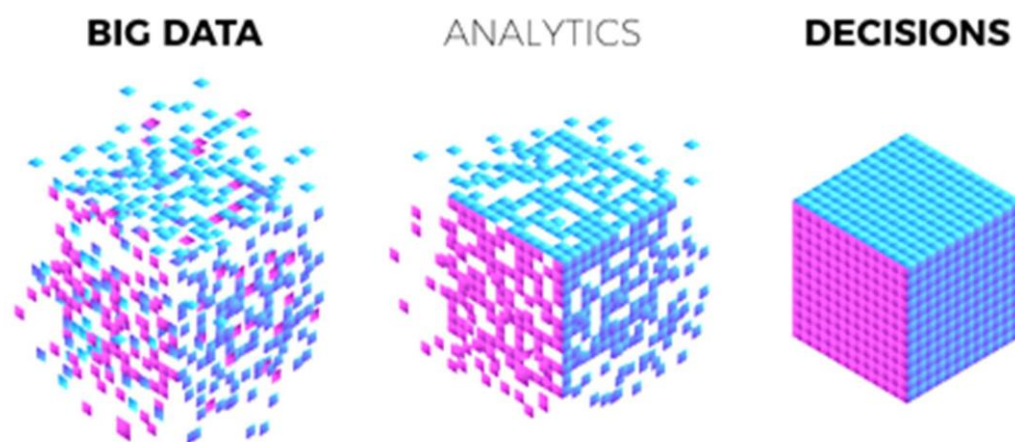
Big data analytics is the often complex process of examining big data to uncover information such as hidden patterns, correlations, market trends and customer preferences that can help organizations make informed business decisions. Big Data analytics deals with collection, storage, processing and analysis of this massive scale data. Specialized tools and frameworks are required for big data analysis when:

- ✓ (1) the volume of data involved is so large that it is difficult to store, process and analyse data on a single machine,
- ✓ (2) the velocity of data is very high and the data needs to be analysed in real-time,
- ✓ (3) there is variety of data involved, which can be structured, unstructured or semi-structured, and is collected from multiple data sources,
- ✓ (4) various types of analytics need to be performed to extract value from the data such as descriptive, diagnostic, predictive and prescriptive analytics.

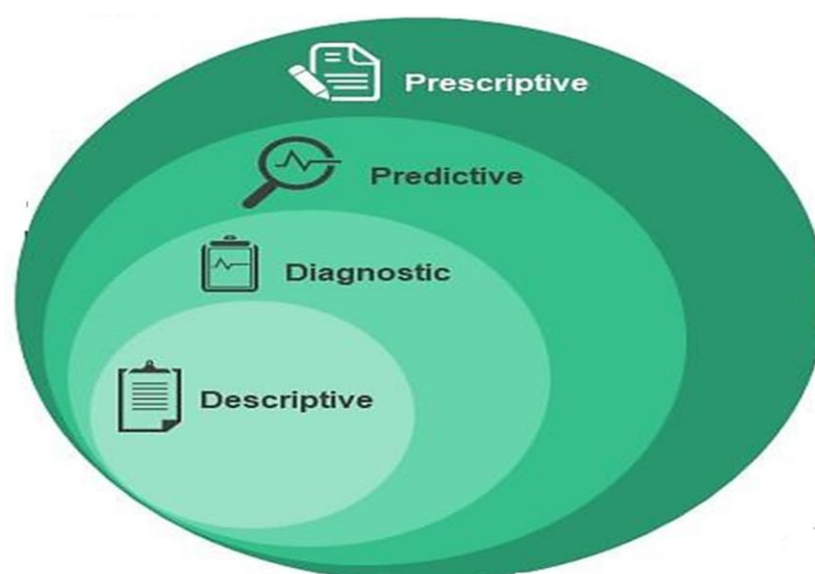
Big data analytics involves several steps starting from data cleansing, data munging (or wrangling), data processing and visualization. Big data analytics life-cycle starts from the collection of data from multiple data sources. Specialized tools and frameworks are required to ingest the data from different sources into the big data analytics backend. The data is stored in specialized storage solutions (such as distributed file systems and non-relational databases) which are designed to scale.

Based on the analysis requirements (batch or real-time), and type of analysis to be performed (descriptive, diagnostic, predictive, or predictive) specialized frameworks are used. Big data analytics is enabled by several technologies such as cloud computing, distributed and parallel processing frameworks, non-relational databases, in-memory computing, for instance. Some examples of big data are listed as follows:

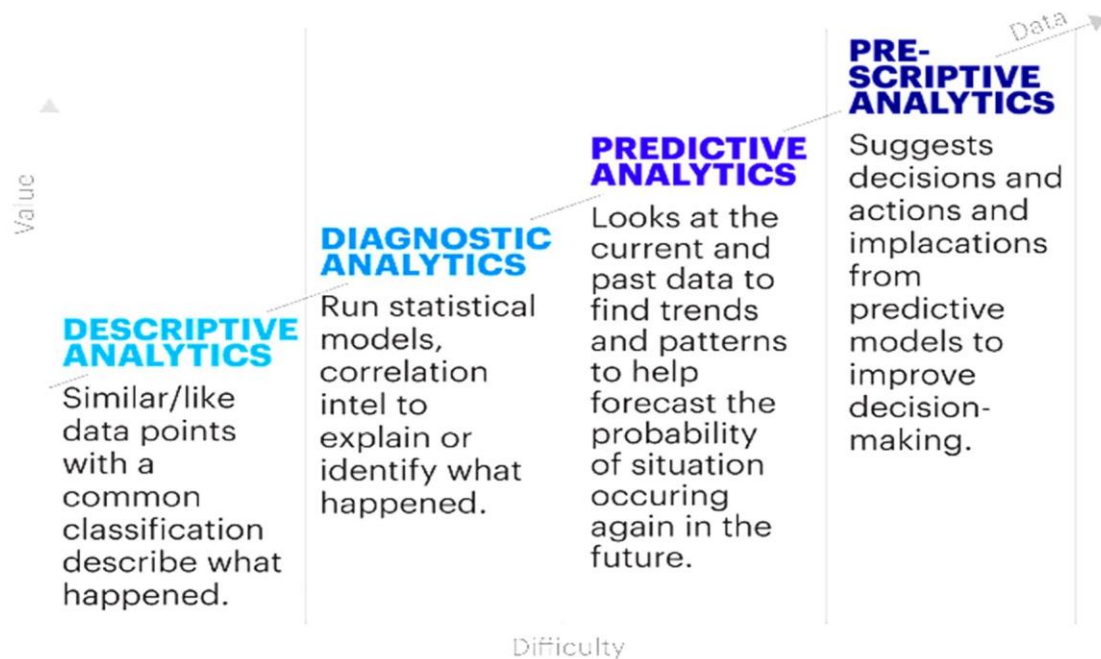
- ✓ Data generated by social networks including text, images, audio and video data
- ✓ Click-stream data generated by web applications such as e-Commerce to analyse user behaviour
- ✓ Machine sensor data collected from sensors embedded in industrial and energy systems for monitoring their health and detecting failures
- ✓ Healthcare data collected in electronic health record (EHR) systems
- ✓ Logs generated by web applications
- ✓ Stock markets data
- ✓ Transactional data generated by banking and financial applications



❖ Types of Analytics



- **Descriptive Analytics :-** Descriptive Analytics is the examination of data or content, usually manually performed, to answer the question “What happened?” (or What is happening?), characterized by traditional business intelligence (BI) and visualizations such as pie charts, bar charts, line graphs, tables, or generated narratives. Descriptive analytics is the interpretation of historical data to better understand changes that have occurred in a business. For example, computing the total number of likes for a particular post, computing the average monthly rainfall or finding the average number of visitors per month on a website.
- **Diagnostic Analytics:-** Diagnostic analytics is a form of advanced analytics that examines data or content to answer the question, “Why did it happen?” It is characterized by techniques such as drill-down, data discovery, data mining and correlations. Diagnostic analysis can be done manually, using an algorithm, or with statistical software (such as Microsoft Excel). For example, a system that collects and analyses sensor data from machines for monitoring their health and predicting failures. Diagnostic analytics can provide more insights into why certain a fault has occurred based on the patterns in the sensor data for previous faults.
- **Predictive Analytics:-** Predictive analytics is a branch of advanced analytics that makes predictions about future outcomes using historical data combined with statistical modelling, data mining techniques and machine learning. Predictive analytics aims to answer - What is likely to happen?. Predictive Analytics is done using predictive models which are trained by existing data. These models learn patterns and trends from the existing data and predict the occurrence of an event or the likely outcome of an event or forecast numbers. Developing prediction models is to divide the existing data into training and test data sets (for example 75% of the data is used for training and 25% data is used for testing the prediction model). For example, Predictive analytics are used to determine customer responses or purchases, as well as promote cross-sell opportunities.
- ❖ **Prescriptive Analytics:-** Prescriptive analytics is a type of data analytics the use of technology to help businesses make better decisions through the analysis of raw data. Specifically, prescriptive analytics factors information about possible situations or scenarios, available resources, past performance, and current performance, and suggests a course of action or strategy. Prescriptive analytics aims to answer - What can we do to make it happen? Prescriptive Analytics can predict the possible outcomes based on the current choice of actions. For example, prescriptive analytics can be used to prescribe the best medicine for treatment of a patient based on the outcomes of various medicines for similar patients. Another example of prescriptive analytics would be to suggest the best mobile data plan for a customer based on the customer’s browsing patterns.



❖ Big data analytics advantages and disadvantages

✓ Benefits or advantages of Big Data

Following are the benefits or advantages of Big Data:

- ➡ Big data analysis derives innovative solutions. Big data analysis helps in understanding and targeting customers. It helps in optimizing business processes.
- ➡ It helps in improving science and research.
- ➡ It improves healthcare and public health with availability of record of patients.
- ➡ It helps in financial trading's, sports, polling, security/law enforcement etc.
- ➡ Any one can access vast information via surveys and deliver answers of any query.
- ➡ Every second additions are made.
- ➡ One platform carry unlimited information.

✓ Drawbacks or disadvantages of Big Data

Following are the drawbacks or disadvantages of Big Data:

- ➡ Traditional storage can cost lot of money to store big data.
- ➡ Lots of big data is unstructured.
- ➡ Big data analysis violates principles of privacy.
- ➡ It can be used for manipulation of customer records.
- ➡ It may increase social stratification.
- ➡ Big data analysis is not useful in short run. It needs to be analyzed for longer duration to leverage its benefits.
- ➡ Big data analysis results are misleading sometimes.
- ➡ Speedy updates in big data can mismatch real figures.

❖ Challenges of conventional systems

The challenges in Big Data are the real implementation hurdles. These require immediate attention and need to be handled because if not handled then the failure of the technology may take place which can also lead to some unpleasant result.

- *Sharing and Accessing Data:*

- Perhaps the most frequent challenge in big data efforts is the inaccessibility of data sets from external sources.
- Sharing data can cause substantial challenges.
- It include the need for inter and intra- institutional legal documents.
- Accessing data from public repositories leads to multiple difficulties.
- It is necessary for the data to be available in an accurate, complete and timely manner because if data in the companies information system is to be used to make accurate decisions in time then it becomes necessary for data to be available in this manner.

- *Privacy and Security:*

- It is another most important challenge with Big Data. This challenge includes sensitive, conceptual, technical as well as legal significance.
- Most of the organizations are unable to maintain regular checks due to large amounts of data generation. However, it should be necessary to perform security checks and observation in real time because it is most beneficial.
- There is some information of a person which when combined with external large data may lead to some facts of a person which may be secretive and he might not want the owner to know this information about that person.
- Some of the organization collects information of the people in order to add value to their business. This is done by making insights into their lives that they're unaware of.

- *Analytical Challenges:*

- There are some huge analytical challenges in big data which arise some main challenges questions like how to deal with a problem if data volume gets too large?
- Or how to find out the important data points?
- Or how to use data to the best advantage?
- These large amount of data on which these type of analysis is to be done can be structured (organized data), semi-structured (Semi-organized data) or unstructured (unorganized data). There are two techniques through which decision making can be done:
 - Either incorporate massive data volumes in the analysis.
 - Or determine upfront which Big data is relevant.

- *Technical challenges:*

- Quality of data:
 - When there is a collection of a large amount of data and storage of this data, it comes at a cost. Big companies, business leaders and IT leaders always want large data storage.
 - For better results and conclusions, Big data rather than having irrelevant data, focuses on quality data storage.
 - This further arise a question that how it can be ensured that data is relevant, how much data would be enough for decision making and whether the stored data is accurate or not.
- Fault tolerance:

- Fault tolerance is another technical challenge and fault tolerance computing is extremely hard, involving intricate algorithms.
- Nowadays some of the new technologies like cloud computing and big data always intended that whenever the failure occurs the damage done should be within the acceptable threshold that is the whole task should not begin from the scratch.
- Scalability:
 - Big data projects can grow and evolve rapidly. The scalability issue of Big Data has lead towards cloud computing.
 - It leads to various challenges like how to run and execute various jobs so that goal of each workload can be achieved cost-effectively.
 - It also requires dealing with the system failures in an efficient manner. This leads to a big question again that what kinds of storage devices are to be used.

❖ **Intelligent Data Analysis (IDA)**

- Intelligent data analysis discloses hidden facts that are not known previously and provides potentially important information or facts from large quantities of data (White, 2008). It also helps in making a decision. Based on machine learning, artificial intelligence, recognition of pattern, and records and visualization technology mainly, IDA helps to obtain useful information, necessary data and interesting models from a lot of data available online in order to make the right choices.
- Intelligent data analysis helps to solve a problem that is already solved as a matter of routine. If the data is collected for the past cases together with the result that was finally achieved, such data can be used to revise and optimize the presently used strategy to arrive at a conclusion. In certain cases, if some questions arise for the first time, and have only a little knowledge about it, data from the related situations helps us to solve the new problem or any unknown relationships can be discovered from the data to gain knowledge in an unfamiliar area.
- Intelligent Data Analysis provides a forum for the examination of issues related to the research and applications of Artificial Intelligence techniques in data analysis across a variety of disciplines. These techniques include (but are not limited to): all areas of data visualization, data pre-processing (fusion, editing, transformation, filtering, sampling), data engineering, database mining techniques, tools and applications, use of domain knowledge in data analysis, big data applications, evolutionary algorithms, machine learning, neural nets, fuzzy logic, statistical pattern recognition, knowledge filtering, and post-processing.

➤ **Steps Involved In IDA:**

IDA, in general, includes three stages: (1) Preparation of data; (2) data mining; (3) data validation and explanation . The preparation of data involves opting for the required data from the related data source and incorporating it into a data set that can be used for data mining. The main goal of intelligent data analysis is to obtain knowledge. Data analysis is the process of a combination of extracting data from data set, analyzing, classification of data, organizing, reasoning, and so on. It is challenging to choose suitable methods to resolve the complexity of the process. Regarding the term visualization, we have moved away from visualization to use the term charting. The term analysis is used for the method of incorporating, influencing, filtering and scrubbing the data, which certainly contains, but is not limited to interrelating with their data through charts.

❖ **Nature of Data**

➤ **What is Data?**

Data is a collection of information gathered by observations, measurements, research or analysis. They may consist of facts, numbers, names, figures or even description of things. Data is organized in the form of graphs, charts or tables. There exist data scientist who does data mining and with the help of that data analyse our world. In general, data is a distinct piece of information that is gathered and translated for some purpose. If data is not formatted in a specific way, it does not valuable to computers or humans. Data can be available in terms of different forms, such as bits and bytes stored in electronic memory, numbers or text on pieces of paper, or facts stored in a person's mind.

➤ **Properties of data**

- 1. Consistency

The element of consistency removes room for contradictory data. Rules will have to be set around consistency metrics, which include range, variance, and standard deviation.

- 2. Accuracy

It is a necessity for DQ data to remain error-free and precise, which means it should be free of erroneous information, redundancy, and typing errors. Error ratio and deviation are two examples of accuracy metrics.

- 3. Completeness

The data should be complete without any missing data. To deliver cloud data quality tools, all data entries should be complete with no room for lapses. The completeness metric is defined as the percentage of complete data records.

- 4. Auditability

The ability to trace data and analyses the changes over time adds to the Data Quality dimensions of audibility of data. An example of audacity metrics is the percentage of the gaps in data sets, modified data, and untraceable and disconnected data.

- 5. Validity

Quality data in terms of validity indicates that all data is aligned with the existing formatting rules. An example of a validity metric is the percentage of data records in the required format.

- 6. Uniqueness

There will be no overlapping of data and it will be recorded only once. The same data may be used in multiple ways, but it will remain unique. Uniqueness metrics are defined by the percentage of repeated values.

- 7. Timeliness

For data to retain its quality, it should be recorded promptly to manage changes. Weekly over annually, tracking is the solution to timeliness. An example of timeliness metrics is time variance.

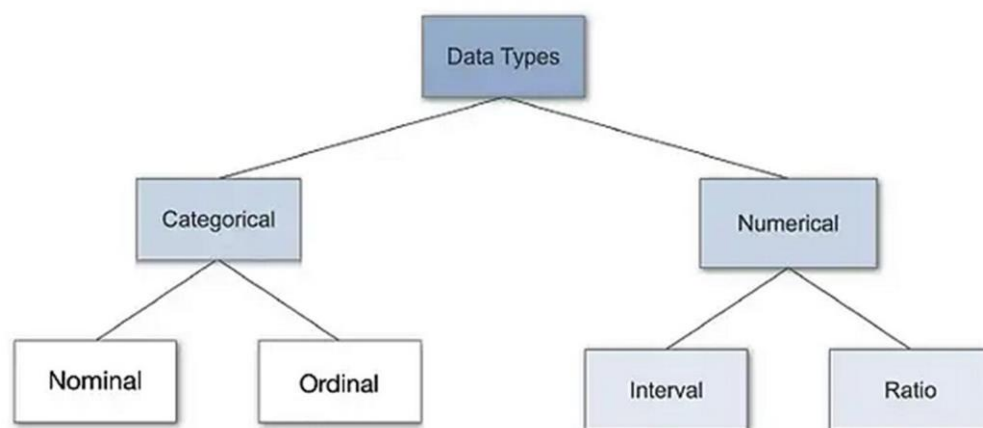
- 8. Relevance

Data captured should be relevant to the purposes for which it is to be used. This will require a periodic review of requirements to reflect changing needs.

- 9. Reliability

Data should reflect stable and consistent data collection processes across collection points and over time. Progress toward performance targets should reflect real changes rather than variations in data collection approaches or methods. Source data is clearly identified and readily available from manual, automated, or other systems and records.

❖ Types of data



➤ Categorical Data

Categorical data represents characteristics. Therefore it can represent things like a person's gender, language etc. Categorical data can also take on numerical values (Example: 1 for female and 0 for male). Note that those numbers don't have mathematical meaning. Categorical data represents characteristics. It is also called qualitative data, which means you cannot express it as a numerical value. Therefore, you cannot measure it. It involves variables such as words, symbols, pictures, which help you sort information based on a category for example, holiday destinations, gender, language, etc.

➤ Nominal Data

Nominal values represent discrete units and are used to label variables, that have no quantitative value. Just think of them as „labels“. Note that nominal data that has no order. Therefore if you would change the order of its values, the meaning would not change. Nominal data are used to label variables where there is no quantitative value and has no order. So, if you change the order of the value then the meaning will remain the same.

Thus, nominal data are observed but not measured, are unordered but non-equidistant, and have no meaningful zero. The only numerical activities you can perform on nominal data is to state that perception is (or isn't) equivalent to another (equity or inequity), and you can use this data to amass them.

➤ *You can't organize nominal data, so you can't sort them.*

Neither would you be able to do any numerical tasks as they are saved for numerical data. With nominal data, you can calculate frequencies, proportions, percentages, and central points. Examples of Nominal data:

✓ What languages do you speak?

- English
- German
- French
- Punjabi

✓ What's your nationality?

- American
- Indian
- Japanese
- German

➤ **Ordinal Data**

Ordinal values represent discrete and ordered units. It is therefore nearly the same as nominal data, except that it's ordering matters. Ordinal data is almost the same as nominal data but not in the case of order as their categories can be ordered like 1st, 2nd, etc. However, there is no continuity in the relative distances between adjacent categories. Examples of Ordinal data:

✓ Opinion

- Agree
- Disagree
- Mostly agree
- Neutral
- Mostly disagree

✓ Time of day

- Morning
- Noon
- Night

➤ **Numerical Data**

This data type tries to quantify things and it does by considering numerical values that make it countable in nature. The price of a smartphone, discount offered, number of ratings on a product, the frequency of processor of a smartphone, or ram of that particular phone, all these things fall under the category of Quantitative data types. It is one of the simplest data types to understand. As the name says, it represents a numerical value and helps answer questions like how many, how much, how long, etc. For example,

- ✓ The number of apples
- ✓ The number of students in a class
- ✓ The height/weight of a person
- ✓ It attempts to quantify items by measuring numerical variables that make them count in nature. The key here is that a numerical variable can take an infinite number of values.

For example, the height of a person can vary from x cm to y cm and can be further broken down based on the fractional values.

➤ **Interval Data**

Interval Data are measured and ordered with the nearest items but have no meaningful zero. The central point of an Interval scale is that the word 'Interval' signifies 'space in between', which is the significant thing to recall, interval scales not only educate us about the order but additionally about the value between every item.

Interval data can be negative, though ratio data can't.

Even though interval data can show up fundamentally the same as ratio data, the thing that matters is in their characterized zero-points. If the zero-point of the scale has been picked subjectively, at that point the data can't be ratio data and should be interval data. Examples of Interval data:

- ✓ Temperature (°C or F, but not Kelvin)
- ✓ Dates (1066, 1492, 1776, etc.)
- ✓ Time interval on a 12-hour clock (6 am, 6 pm)

➤ **Ratio Data**

Ratio Data are measured and ordered with equidistant items and a meaningful zero and never be negative like interval data. An outstanding example of ratio data is the measurement of heights. It could be measured in centimetres, inches, meters, or feet and it is not practicable to have a negative height. Ratio data enlightens us regarding the order for variables, the contrasts among them, and they have absolutely zero. It permits a wide range of estimations and surmisings to be performed and drawn. Example of Ratio data:

- ✓ Age (from 0 years to 100+)
- ✓ Temperature (in Kelvin, but not °C or F)
- ✓ Distance (measured with a ruler or any other assessing device)
- ✓ Time interval (measured with a stop-watch or similar)

➤ **Sources and nature of data**

In general, data is any set of characters that is gathered and translated for some purpose, usually analysis. If data is not put into context, it doesn't do anything to a human or computer. There are multiple types of data. Some of the more common types of data include the following:

- ✓ Single character
- ✓ Boolean (true or false)
- ✓ Text (string)

- ✓ Number (integer or floating-point)
- ✓ Picture
- ✓ Sound
- ✓ Video

In a computer's storage, digital data is a sequence of bits (binary digits) that have the value one or zero. Data is processed by the CPU, which uses logical operations to produce new data (output) from source data (input).

➤ What is the difference between data and information?

While data is a collection of individual statistics or facts, information is knowledge gained through research, study, instruction, or communication. A key difference between the two is that data can be unorganized, unrelated, or raw, while information is organized. Data on its own might not have any meaning and might need to be sorted, analyzed, or interpreted to become information. When trying to make a decision, data might not be enough, but decisions can be made based solely on information. Businesses that can collect and use data can gain valuable information from it that will help them make faster and smarter business decisions. Computers are an extremely useful tool to turn data into information using software applications, formulas, and scripts.

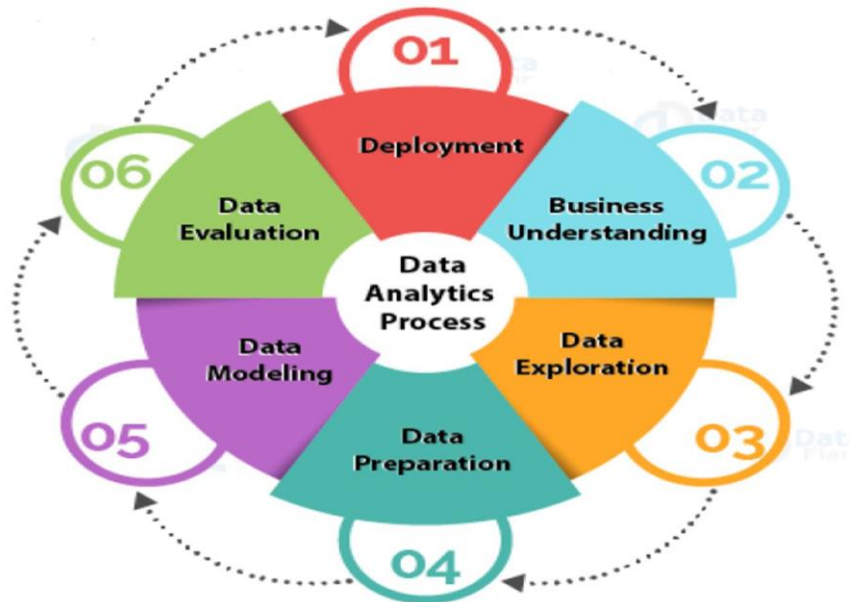
➤ Machine-readable vs. human-readable data

All data can be categorized as machine-readable, human-readable, or both. Human-readable data utilizes natural language formats (such as a text file containing ASCII codes or PDF document), whereas machine-readable data uses formally structured computer languages (Parquet, Avro, etc.) to be read by computer systems or software. Some data is readable by both machines and humans, as in the case of CSV, HTML, or JSON. The line between machine- and human-readable data is becoming increasingly blurred because so many formats that are prevalent today are accessible enough to be navigated by a human yet structured enough to be processed by a machine. This is largely the result of artificial intelligence, machine learning, and automation, which streamlines tasks and workflows so manual data entry and analysis is done by a machine rather than a human. However, these processes need to maintain their human readability in case the programming needs to be adjusted. Most data in these cases also exist in a vacuum and does not have much meaning without context from a human perspective.

❖ Analytic processes

There are 6 analytic processes:

- ✓ 1. Deployment
- ✓ 2. Business Understanding
- ✓ 3. Data Exploration
- ✓ 4. Data Preparation
- ✓ 5. Data Modeling
- ✓ 6. Data Evaluation



Step 1: Deployment

- Here we need to:
 - plan the deployment and monitoring and maintenance,
 - we need to produce a final report and review the project.
- In this phase,
- we deploy the results of the analysis.
- This is also known as reviewing the project.

Step 2: Business Understanding

- Business Understanding
 - The very first step consists of business understanding.
 - Whenever any requirement occurs, firstly we need to determine the business objective,
 - assess the situation,
 - determine data mining goals and then
 - produce the project plan as per the requirement.
- Business objectives are defined in this phase.

Step 3: Data Exploration

- The second step consists of Data understanding.
 - For the further process, we need to gather initial data, describe and explore the data and verify data quality to ensure it contains the data we *require*.
 - Data collected from the various sources is described in terms of its application and the need for the project in this phase.
 - This is also known as data exploration.
- This is necessary to verify the quality of data collected.

Step 4: Data Preparation

- From the *data collected in the last step*,
 - we need to *select data as per the need, clean it, construct it to get useful information* and
 - then *integrate it all*.
- Finally, we need to *format the data to get the appropriate data*.
- Data is selected, cleaned, and integrated into the format finalized for the analysis in this phase.

Step 5: Data Modeling

- we need to
 - select a modeling technique, generate test design, build a model and assess the model built.
- The data model is build to
 - analyze relationships between various selected objects in the data,
- test cases are built for assessing the model and model is tested and implemented on the data in this phase.

❖ Analytic tools

Big data tools for HPC and supercomputing

- MPI (Message Passing Interface)
 - ✓ Provide standardized function interfaces for communication between parallel processes.
- Big data tools on clouds
 - MapReduce model
 - Iterative MapReduce model
 - DAG model
 - Graph model
 - Collective model
- Other BDA tools
 - SaS
 - R
 - Hadoop

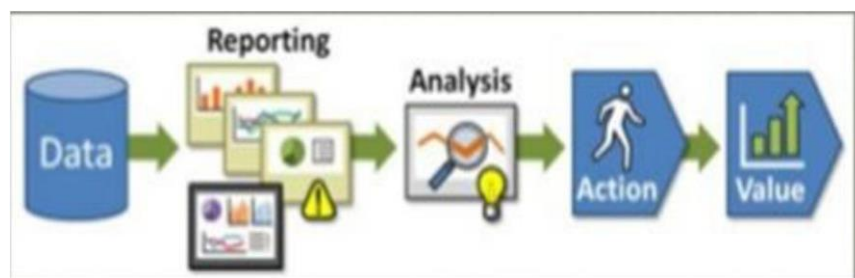
Thus the BDA tools are used through out the BDA applications development.

❖ Analysis vs Reporting

- ❖ Reporting: The process of organizing data into informational summaries in order to monitor how different areas of a business are performing.****
- ❖ Analysis: The process of exploring data and reports in order to extract meaningful insights, which can be used to better understand and improve business performance.
- Data reporting: Gathering data into one place and presenting it in visual representations
- Data analysis: Interpreting your data and giving it context

❖ Comparing analysis with reporting

- ✓ **Reporting** is “the process of organizing data into informational summaries in order to monitor how different areas of a business are performing.”
- Measuring core metrics and presenting them — whether in an email, a slidedeck, or online dashboard — falls under this category.
- ✓ **Analytics** is “the process of exploring data and reports in order to extract meaningful insights, which can be used to better understand and improve business performance.”
- Reporting helps companies to monitor their online business and be alerted to when data falls outside of expected ranges.
- ✓ **Good reporting**
 - should raise questions about the business from its end users.
 - The goal of analysis is
 - to answer questions by interpreting the data at a deeper level and providing actionable recommendations.
- A firm may be focused on the general area of analytics (strategy, implementation, reporting, etc.)
 - but not necessarily on the specific aspect of analysis.
- It's almost like some organizations run out of gas after the initial set-up-related activities and don't make it to the analysis stage



A reporting activity deliberately proposes Analysis activity.

❖ Contrast between analysis and reporting

The basis differences between Analysis and Reporting are as follows:

Analysis	Reporting
Provides what is needed	Provides what is asked for
Is typically customized	Is Typically standardized
Involves a person	Does not involve a person
Is extremely flexible	Is fairly Inflexible

Reporting translates raw data into **information**.

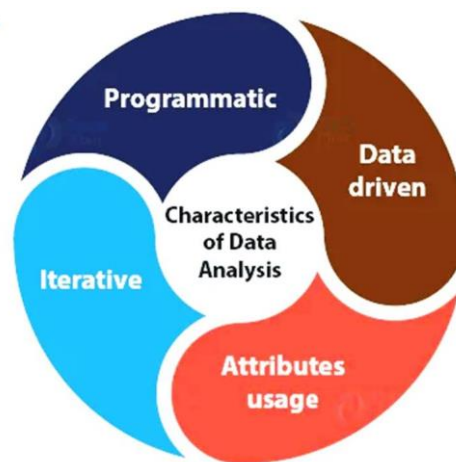
- Analysis transforms data and information into **insights**.
- reporting shows you **what is happening**
- while analysis focuses on explaining **why it is happening** and **what you can do about it**.
- ✓ Reports are like Robots n monitor and alter you and where as analysis is like parents - c an figure out what is going on (hungry, dirty diaper, no pacifier, , teething, tired, ear infection, etc).

- ✓ Reporting and analysis can go hand-in-hand:
- ✓ Reporting provides no limited context about what is happening in the data. Context is critical to good analysis.
- ✓ Reporting translate a raw data into information
- ✓ Reporting usually raises a question – **What is happening ?**
- ✓ Analysis transforms the data into insights - **Why is it happening ? What you can do about it?**

Thus, Analysis and Reporting is synonym to each other with respect their need and utilizing in the needy context.

❖ Characteristics of Data Analysis

The characteristics of the data analysis depend on different aspects such as volume, velocity, and variety.



1. Programmatic

There might be a need to write a program for data analysis by using code to manipulate it or do any kind of exploration because of the scale of the data.

2. Data-driven

A lot of data scientists depend on a hypothesis-driven approach to data analysis. For appropriate data analysis, one can also avail the data to foster analysis. This can be of significant advantage when there is a large amount of data. **For example** machine learning approaches can be used in place of hypothetical analysis.

3. Attributes usage

For proper and accurate analysis of data, it can use a lot of attributes. In the past, analysts dealt with hundreds of attributes or characteristics of the data source. With Big Data, there are now thousands of attributes and millions of observations.

4. Iterative

As whole data is broken into samples and samples are then analyzed, therefore data analytics can be iterative in nature. Better compute power enables iteration of the models until data analysts are satisfied. This has led to the development of new applications designed for addressing analysis requirements and time frames.

❖ Modern Data Analytic Tools

• **Modern Analytic Tools:** Current Analytic tools concentrate on three classes:

- a) Batch processing tools
- b) Stream Processing tools and
- c) Interactive Analysis tools.

a) Big Data Tools Based on Batch Processing:

Batch processing system :-

- Batch Processing System involves
 - collecting a series of processing jobs and carrying them out periodically as a group (or batch) of jobs.
 - It allows a large volume of jobs to be processed at the same time.
 - An organization can schedule batch processing for a time when there is little activity on their computer systems, for example overnight or at weekends.
- One of the **most famous and powerful batch process-based Big Data tools is Apache Hadoop.**
 - ✓ It provides infrastructures and platforms for other specific Big Data applications.

b) Stream Processing tools

- Stream processing – Envisioning (predicting) the life in data as and when it transpires.
- The key strength of stream processing is that **it can provide insights faster, often within milliseconds to seconds.**
 - It **helps understanding the hidden patterns in millions of data records in real time.**
 - It **translates into processing of data from single or multiple sources**
 - in real or near-real time applying the desired business logic and emitting the processed information to the sink.
- Stream processing serves
 - multiple
 - resolves in today's business arena.

Real time data streaming tools are:

a) Storm

- Storm is a **stream processing engine without batch support,**
- a true **real-time processing framework,**
- taking in a stream as an entire 'event' instead of series of small batches.
- *Apache Storm is a distributed real-time computation system.*
- *It's applications are designed as directed acyclic graphs.*

b) Apache flink

- Apache flink is
 - an open source platform
 - which is a **streaming data flow engine** that **provides communication fault tolerance** and
 - **data distribution computation over data stream** .
 - flink is a top level project of Apache flink is scalable data analytics framework that is fully compatible to hadoop .
 - flink can **execute both stream processing and batch processing easily**.
 - flink was **designed as an alternative to map-reduce**.

c) Kinesis

- Kinesis as an out of the box **streaming data tool**.
- Kinesis **comprises of shards which Kafka calls partitions**.
- For organizations that take **advantage of real-time or near real-time access to large stores of data**,
- Amazon Kinesis is great.
- Kinesis Streams solves a variety of streaming data problems.
- One common use is **the real-time aggregation of data which is followed by loading the aggregate data into a data warehouse**.
- Data is put into Kinesis streams.
- This ensures durability and elasticity.

c) Interactive Analysis -Big Data Tools

- The **interactive analysis presents**
 - the **data in an interactive environment**,
 - allowing users to undertake their own analysis of information.
- **Users are directly connected to**
 - the computer and hence can interact with it in real time.
- **The data can be :**
 - reviewed,
 - compared and
 - analyzed
- ***in tabular or graphic format or both at the same time.***